# Elecraft KPA1500 for Zeus

Version 1.1.9 · by KW4EX · Read Me First

**Written for radio operators, not programmers.** You don't need to know anything about programming. Every step tells you exactly what to click. You won't need to install any other software on any computer: everything the add-on needs is in this folder or already inside Zeus.


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**No warranty — use at your own risk.** This add-on is shared free by one radio operator, “as is”. There is no warranty of any kind: no promise that it works, that it is safe, or that it suits your equipment or purpose. Amplifiers, radios and antennas can be damaged, and RF can be dangerous. You are responsible for checking that it behaves correctly with your own station, and for operating within your license. The author is not liable for any damage or loss. It is not made, approved or supported by Elecraft, FlexRadio or the Zeus project.

## 1. What it is

**Zeus** is the SDR program that runs your radio. Zeus can take add-ons, which it calls **community features**. This folder holds an add-on that lets Zeus watch and control an **Elecraft KPA1500** amplifier, including its built-in antenna tuner (ATU).

Once it's installed, Zeus shows a **KPA1500 panel** with the amplifier's meters and buttons, and the amplifier follows your radio's frequency automatically. It does the same jobs as the earlier KPA1500 Thetis bridge, but runs inside Zeus, so you don't need a separate program.

The add-on talks to the amplifier over your home network (Ethernet, UDP port 1500). Because of that, other programs on your network can still talk to the amplifier at the same time, such as Elecraft's own utility or a logger. A USB serial cable, or a "virtual serial port" program, also works for stations set up that way.

The add-on never keys your transmitter. It changes only the amplifier's settings (and, if you ask it to, it can scan bands with the receiver).

## 2. What it does

### Shows (you choose which)

- Forward power, SWR, reflected power and **drive** (input) power as bar meters. During a TUNE, when the amp is bypassed, the panel shows the power passing through, marked "(radio)" or "(bypass)".
- Temperature, PA voltage, PA current, fan speed, ATU status and bypass SWR.
- **The amp's own frequency reading**, all the time. It turns amber if the amp is more than 10 kHz away from your VFO.
- **Receive level** in dBm, used for the "No antenna?" warning.
- Faults, in plain words, with a **Clear fault** button.

### Controls (you choose which, and their order)

- `POWER` on/off, including **power-on over the network** (Wake-on-LAN).
- `OPERATE` / Standby.
- `TUNE STBY`: an on/off switch for Standby on Tune. While it's lit, pressing TUNE in Zeus drops the amp to STANDBY, and OPERATE comes back when TUNE ends. There are no pop-ups.
- `ANT 1` / `ANT 2`, `ATU` in-line / bypass, `ATU TUNE` (full tuner search).
- Band buttons (160–6 m), and the fan's minimum speed.

### Works for you in the background

- **Amp follows the VFO.** When you change frequency or band in Zeus, the amplifier follows within a second, and its ATU recalls the tuning for that spot. With split, it follows the transmit VFO.
- **Standby on Tune**, automatically, whenever you press TUNE in Zeus.
- **"No antenna?" warning.** If the receive level drops below a limit for the band, which usually means no antenna is connected, a warning appears. You can also have it hold the amp in STANDBY. You set the limits, or let it learn them for each band.
- **Find amplifier.** It finds the KPA1500 on your network, so you don't need to know its IP address.
- **Safety locks.** Antenna, band and ATU changes are refused while transmitting. Going to STANDBY is always allowed.

## 3. What you need

| Item | Details |
|---|---|
| **Zeus** | Installed and working with your radio, on a version that has the **Features** menu with a **Community** section. If you can't find that menu, update Zeus first. |
| **Elecraft KPA1500** | Connected to the **same home network** as the Zeus computer (recommended), *or* to it by its rear **USB** cable. |
| **A computer running Zeus** | Any one of: a **Windows** 10 or 11 PC (64-bit) · a **Mac** with Apple silicon (M1–M4) or Intel · a **Linux PC** (64-bit) · a **Raspberry Pi 4 or 5** running the **64-bit** Raspberry Pi OS. |
| **Other software** | **None.** You don't need .NET, Python or drivers. A Windows PC may need the FTDI USB driver for the serial cable; Windows usually installs it automatically. |
| **Optional: Zeus TCI** | A setting inside Zeus, **off** when Zeus is new. Turn it on to get: Standby on Tune, following split, the receive level for the "No antenna?" warning, and drive power during TUNE. Part 9 shows how. |
| **Optional: amp setting** | For power-on over the network, the KPA1500's menu item **NET WAKE on LAN** must be ON. The panel can turn it on for you. |

## 4. What's in this folder

| Folder or file | What it is |
|---|---|
| **READ ME FIRST ...pdf** | This guide. |
| **1 Windows PC** | The install file for Windows. |
| **2 Mac - Apple silicon (M1-M4)** | The install file for newer Macs. |
| **3 Mac - Intel** | The install file for older Intel Macs. |
| **4 Linux PC (64-bit)** | The install file for an ordinary Linux PC. |
| **5 Raspberry Pi (64-bit)** | The install file for a Raspberry Pi 4 or 5 (and other 64-bit ARM Linux). |

Each numbered folder holds one file ending in `.zip`. That `.zip` file **is the add-on**. Zeus opens it itself.

> **Do not unzip (extract) the .zip file.** Give it to Zeus exactly as it is. If your computer unzips it by accident and you end up with a folder, ignore the folder and use the original `.zip`.

**Getting the file to another computer:** copy the right numbered folder to a USB stick, or across your home network, onto the Mac or Linux computer. Or download it there from `github.com/kw4ex/zeus-amp/tree/builds/latest`: click the file name, then the **download** button (↓, "Download raw file").

## 5. Which file is mine?

| Computer that runs Zeus | How to tell | Use the folder |
|---|---|---|
| Windows PC | It runs Windows. | **1 Windows PC** |
| Mac, Apple silicon | Apple menu › **About This Mac** › "Chip" says **Apple M1, M2, M3 or M4** | **2 Mac - Apple silicon** |
| Mac, Intel | Apple menu › **About This Mac** › "Processor" says **Intel** | **3 Mac - Intel** |
| Linux PC | An ordinary desktop or laptop running Linux (Ubuntu, Mint, Debian, Fedora ...) | **4 Linux PC** |
| Raspberry Pi | Open **Terminal**, type `uname -m`, press Enter. It must say **aarch64**. If it says `armv7l`, the Pi is running the 32-bit system, which this add-on (and Zeus) can't use; you'd need to install the 64-bit Raspberry Pi OS. | **5 Raspberry Pi** |

The file names end the same way: `…-1.1.9.zip` (Windows), `…-osx-arm64.zip`, `…-osx-x64.zip`, `…-linux-x64.zip`, `…-linux-arm64.zip`.

## 6. Install on a Windows PC

1. **Close the Thetis bridge** (and any other program that *controls* the amp) whenever Zeus is running. Two programs steering one amplifier fight each other, which shows up as odd behaviour. Programs that only *read* the amp, such as a logger, are fine.
2. Start **Zeus**.
3. You don't need to remove an older version first. Zeus has no Remove button, and installing a new version replaces the old one.
4. Open **Features**, choose the **Community** tab, and click **Install local feature**.
5. In the window that opens, browse to **C: › RadioScripts › KPA1500_Zeus_Plugin_v1.1.9 › 1 Windows PC**, click the `.zip` file once, then click **Open**.
6. Zeus lists **Elecraft KPA1500** as installed. If it asks to restart, close Zeus and start it again.
7. Add the panel: use Zeus's **Add Panel** option, open **Amplifiers**, choose **KPA1500**. Put the panel wherever you like on screen.
8. If Windows shows a **Windows Security / Firewall** message about Zeus, tick **Private networks** and click **Allow**. Zeus needs this to reach the amp.
9. Check that the panel's top line says **KW4EX v1.1.9**. Then go to **Part 9**.

The menu names come from Zeus. If your version words them slightly differently, look for the closest match.

## 7. Install on a Mac

1. Copy the folder **2 Mac - Apple silicon** or **3 Mac - Intel** (Part 5 tells you which) onto the Mac, for example into **Downloads**. Or download the file from GitHub (Part 4).
2. If Safari unzips the file into a folder by itself, that's fine. Just make sure you still have the `.zip` file itself.
3. Start **Zeus**. Close the Thetis bridge or any other program that controls the amp. Installing a new version replaces an older one.
4. Open **Features** › **Community** › **Install local feature**.
5. Pick the `.zip` file (`…-osx-arm64.zip` or `…-osx-x64.zip`) and click **Open**.
6. If Zeus asks to restart, quit it with **Command-Q** and start it again.
7. Add the panel: **Add Panel › Amplifiers › KPA1500**.
8. **Local network permission:** if the Mac asks to let Zeus "find and connect to devices on your local network", click **Allow**. If you clicked Don't Allow by mistake, open **System Settings › Privacy & Security › Local Network** and switch **Zeus** on.
9. Check that the top line says **KW4EX v1.1.9**, then go to **Part 9**.

**USB cable on a Mac:** in the panel's Settings, the amp shows up as a port starting `/dev/cu.usbserial-`.

## 8. Install on Linux or a Raspberry Pi

1. Copy the folder **4 Linux PC** or **5 Raspberry Pi** onto the computer, for example into your home folder or **Downloads**. Or download the file from GitHub (Part 4).
2. Start **Zeus**. Close the Thetis bridge or any other program that controls the amp. Installing a new version replaces an older one.
3. Open **Features** › **Community** › **Install local feature**.
4. Pick the `.zip` file (`…-linux-x64.zip` for a PC, `…-linux-arm64.zip` for a Pi) and click **Open**.
5. If Zeus asks to restart, close it and start it again.
6. Add the panel: **Add Panel › Amplifiers › KPA1500**.
7. Check that the top line says **KW4EX v1.1.9**, then go to **Part 9**.

**Only if you use the USB cable (skip this for network):** Linux lets only members of the **dialout** group use serial ports. Do this once:

1. Open **Terminal**: on a Pi, click the black screen icon at the top; on most Linux PCs, press **Ctrl + Alt + T**.
2. Type exactly this line and press **Enter**: `sudo usermod -aG dialout $USER`
3. Type your computer password when asked. Nothing shows while you type; that's normal. Press **Enter**.
4. **Log out and back in**, or restart the computer.

The amp then appears in the panel's Settings as `/dev/ttyUSB0` (or a similar name).

## 9. First-time settings (all computers)

Click the ⚙ **gear** at the top right of the KPA1500 panel to open Settings. After you change anything, scroll down and click **Save**.

### A. Connect to the amplifier: network (recommended)

1. Make sure the KPA1500 is turned on and plugged into your home network.
2. Under **Connection**, leave **Network (UDP)** selected.
3. Click **Find amplifier**. Within a few seconds it lists your KPA1500 with its address, serial number and firmware.
4. Click **Use** next to it. The address fills in for you. If you already know it (it's on the amp's NET menu), you can type it instead.
5. Leave **UDP port** at **1500**. Click **Save**.
6. Within a few seconds the POWER button lights green and the meters come alive.

### B. Or connect by USB cable / virtual serial port

1. Under **Connection**, choose **Serial port (USB or virtual)** and click **Refresh**.
2. Pick the amp's port: `COM3` or similar on Windows, `/dev/cu.usbserial-…` on a Mac, `/dev/ttyUSB0` on Linux. A virtual serial port program shows the port name you gave it.
3. Set **Baud** to match the amp's **SERIAL SPEED HOST** menu. Click **Save**.

Only one program at a time can use a serial port. Over the network there's no such limit.

### C. Power-on over the network (Wake-on-LAN)

- The panel **learns the amp's network ID (MAC address) by itself** the first time the amp is on and connected. After that, POWER can switch the amp on from off.
- The amp's menu item **NET WAKE on LAN** must be ON. If it's off, the panel shows a note with an **Enable Wake-on-LAN** button. Click it once.

### D. Turn on Zeus TCI (for Standby on Tune, split, receive level and drive during TUNE)

1. In **Zeus's own settings** (not the panel's), find **TCI** and switch it **on**. Zeus's TCI port is normally **40001**. Note the number if yours is different.
2. In the panel's Settings, under **Standby on Tune**, tick **Use Zeus TCI …**. Leave **TCI host** as `127.0.0.1`, set **TCI port** to **40001** (or Zeus's number), and leave **TCI receiver** at `0`.
3. Keep **Return to OPERATE afterwards** ticked and click **Save**. Then turn Standby on Tune on with the TUNE STBY button at the top of the panel; it stays lit while it's on. (The **Enable Standby-on-Tune** tick box in Settings does the same thing.)
4. A green **TCI** tag appears on the panel's top line. If it says **TCI WAIT**, the line in Settings tells you why. Usually TCI is still off in Zeus, or the port number doesn't match.

### E. The "No antenna?" warning (optional)

This needs TCI (step D). Under **No antenna? check**:

- Tick **Warn when the receive level says no antenna**. Choose **Warn only**, or **Warn and keep the amp in STANDBY**.
- Each band has a limit in dBm. Below the limit, you get the warning. The easiest choice is to leave **Auto-learn** ticked: the panel learns each band's normal noise while you listen.
- Or set limits yourself, band by band:
  - **Learn this band (antenna connected)**: sets the limit 8 dB below the noise you hear now.
  - **Learn this band (antenna disconnected)**: with the antenna unplugged, sets the limit 8 dB above the radio's own noise.
  - Or type a number into the table. A number you type is never changed automatically. Leave a box empty to switch the check off on that band.
- **Scan all bands now** tunes the receiver to the middle of each band's phone segment, sets every band's limit, then puts your frequency back. It needs Zeus's permission to tune the radio. If that button is greyed out, use the other methods.
- When you use a dummy load on purpose, click **DUMMY OK** on the warning to silence it.

The attenuator, preamp off, a very narrow filter or a dead band at night can all lower the receive level and give a false warning.

### F. Choose and arrange what the panel shows

- **Telemetry shown:** untick any meter or reading you don't want.
- **Controls at the top:** each button has a row. Tick or untick it to show or hide that button. Press ▲ or ▼ to move it up or down; the buttons appear in that order across the top of the panel. **Default order** puts them back.
- **Display: Compact layout** makes the panel smaller. You can also set °C or °F, the meter's full scale, and the SWR and temperature warning points.
- **Behaviour:** choose whether to ask before power off or on, and whether the amp follows the VFO. Following the VFO is on by default and recommended.

## 10. Using the panel

**A lit (coloured) button means "this is on / in use now."** A plain button means off. A faint button is the other choice of a pair. A dashed outline means the button is locked for the moment: the amp is transmitting, the ATU is tuning, or the amp is off.

| Button | Lit means | Click it to |
|---|---|---|
| POWER | Amp is on (green) | Turn the amp off, or on. Power-on works over the network. |
| OPERATE | Amp is in OPERATE (blue). Plain = STANDBY. | Switch between OPERATE and STANDBY. STANDBY always works, even while transmitting. |
| TUNE STBY | Standby on Tune is ON (blue) | Turn Standby on Tune on or off. While it's on, every TUNE in Zeus puts the amp in STANDBY, and OPERATE returns when TUNE ends, with no pop-ups. Needs Zeus TCI (Part 9D). The setting is remembered. |
| ANT 1 ANT 2 | The lit one is in use | Switch antennas. |
| ATU | Tuner in-line. Plain = bypassed. | Put the tuner in-line or bypass it. |
| ATU TUNE | Tuner is searching (amber) | Start a full tuner search. You then send a low-power carrier (TUNE) from the radio. Click again to cancel. |
| Band buttons | The amp's current band | Change the amp's band. Normally you don't need to, because the amp follows the radio. |

- **Top line:** "KW4EX v1.1.9", the amp's address, firmware, band and antenna, plus tags: **TCI** (green when connected to Zeus), **TX** (red while transmitting), **FAULT**, and the ⚙ settings button.
- **Coloured bars under the buttons** are messages, such as a fault (with **Clear fault**), "No antenna?", or Standby on Tune in progress (with **Cancel**).
- **Drive** shows the amp's own input reading in OPERATE. During TUNE, when the amp is bypassed, it shows the power passing through: from the radio over TCI, marked **(radio)**, or else from the amp's own meter, marked **(bypass)**.

## 11. A safe first test

Use a dummy load or a well-matched antenna, and keep the radio at low power (about 10 W) for the transmit steps. Tick each item as it works:

1. With nothing transmitting, the band, antenna, ATU, temperature and volts match the amp's front panel.
2. In Settings, **Find amplifier** lists your KPA1500.
3. Change frequency, then band, in Zeus. The amp follows within a second or two. **Amp freq** matches your VFO.
4. With TCI on, the **RX level** shows a dBm number that changes with band noise.
5. Amp in OPERATE, press **TUNE** in Zeus at about 10 W. OPERATE goes out (STANDBY) and Drive shows about 10 W **(radio)** or **(bypass)**. When TUNE stops, OPERATE lights again by itself.
6. While transmitting, the ANT, ATU and band buttons are locked (dashed outline).
7. Click **ATU TUNE**, then **Start**, then TUNE on the radio at low power. The tuner searches and finishes.
8. POWER off turns the amp off. POWER on wakes it within about 15 seconds.

If anything misbehaves, click **OPERATE** to put the amp in STANDBY, then note which step failed and what you saw.

## 12. Updating to a newer version

1. Each new version comes in its own folder with the version number in its name, for example **KPA1500_Zeus_Plugin_v1.2.0**. Keep the old folder until the new one works.
2. In Zeus: **Features › Community › Install local feature**, and pick the new `.zip`, as in Parts 6–8. It replaces the old version; there's nothing to remove first. Close and restart Zeus afterwards.
3. Check that the panel's top line shows the new version number. Your settings are normally kept. If they aren't, set them again (Part 9).

The newest version is always at `github.com/kw4ex/zeus-amp/tree/builds/latest`.

## 13. If something goes wrong

| What you see | What to do |
|---|---|
| Zeus won't install the file | Pick the `.zip` *file*, not a folder, and the right one for your computer (Part 5). Don't unzip it first. |
| No KPA1500 under Add Panel | Restart Zeus. Check that **Features › Community** lists Elecraft KPA1500 as installed and enabled. |
| Top line shows an older version | Install the new file again, then close and restart Zeus. |
| Odd behaviour after a few minutes: the amp changes frequency or mode by itself, or the network gets sluggish | Two programs are controlling the amp. Close the Thetis bridge (and Thetis). In Zeus's **Features** list, check there is only **one** Elecraft KPA1500 entry. |
| Yellow bar: "Another program is also controlling the amp" | Something else keeps changing the amp's frequency, usually the Thetis bridge. Close it. The panel puts Zeus's frequency back each time. |
| OFFLINE or NO REPLY | Is the amp on? Use **Find amplifier** (Part 9A). On a Mac, allow Local Network (Part 7). On Windows, allow Zeus through the firewall (Part 6). |
| Find amplifier finds nothing | The computer and the amp must be on the same home network. Some Wi-Fi routers block searches; type the amp's IP address instead (from its NET menu). |
| Serial list is empty | Check the USB cable and click **Refresh**. On Linux, do the dialout step (Part 8). On Windows, check Device Manager › Ports for a COM port. |
| Serial port "busy" | Another program has it open. Close it, or use the network connection instead. |
| TCI WAIT, or RX level says "TCI off" | Turn TCI on in Zeus and check the port (40001), Part 9D. |
| TUNE STBY shows ⚠, or TUNE doesn't put the amp in STANDBY | Zeus TCI isn't connected. Turn TCI on in Zeus's own settings (port 40001). In the panel's Settings, the TCI line says what's wrong. Turning TUNE STBY on also switches on the panel's "Use Zeus TCI" box. |
| Drive blank during TUNE | Needs TCI connected (green TCI tag). |
| POWER on doesn't wake the amp | Turn on **NET WAKE on LAN** in the amp, or turn the amp on once from its front panel so the add-on learns its MAC address. |
| "No antenna?" warning when the antenna is fine | Check the attenuator and preamp. Lower that band's limit, or click **Learn this band (antenna connected)**. |

| | |
|---|---|
| "Amp freq ≠ VFO" | In Settings, check that **Amp follows the VFO** is ticked. The amp only follows when it is connected and on. |

**Asking for help:** take a picture of the panel. On Windows press **Windows + Shift + S**; on a Mac press **Command + Shift + 4**; on Linux press **Print Screen**. Ticking **Event log** under Telemetry shown lists every command the add-on sends, which helps find problems. Send the picture with a short note of what you did and what happened.

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